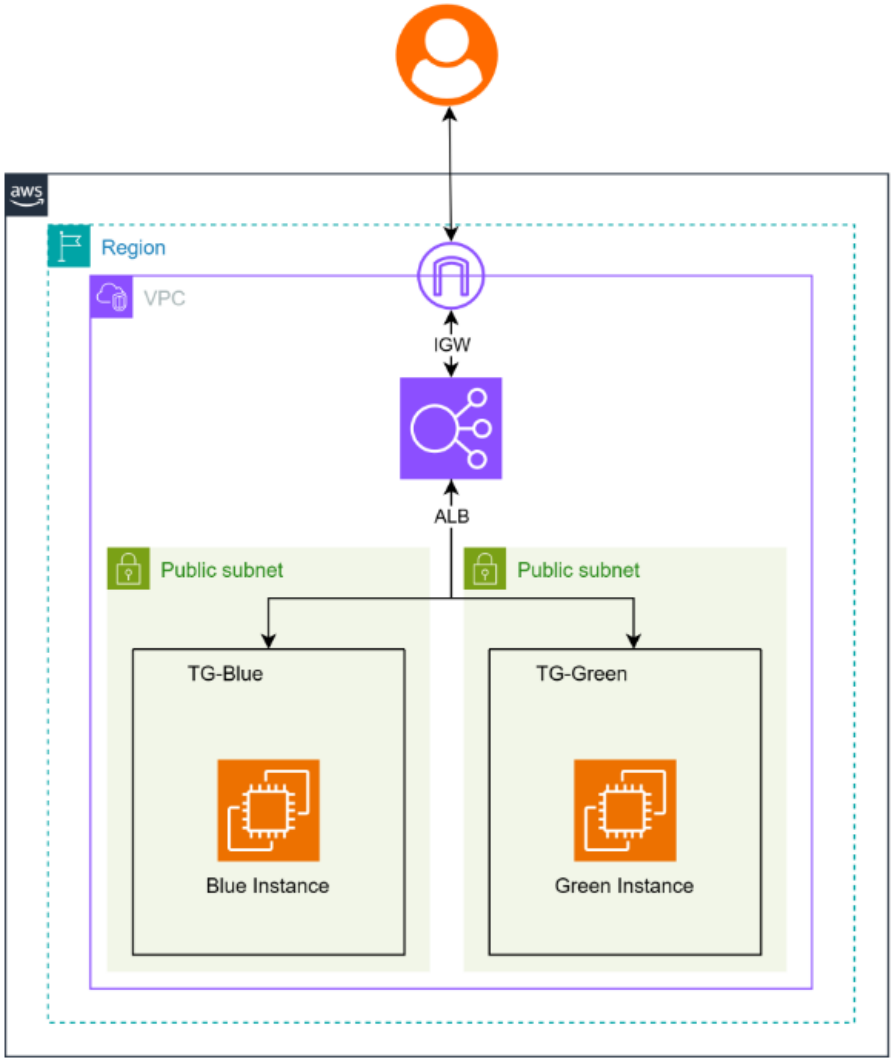
Blue-Green Deployment

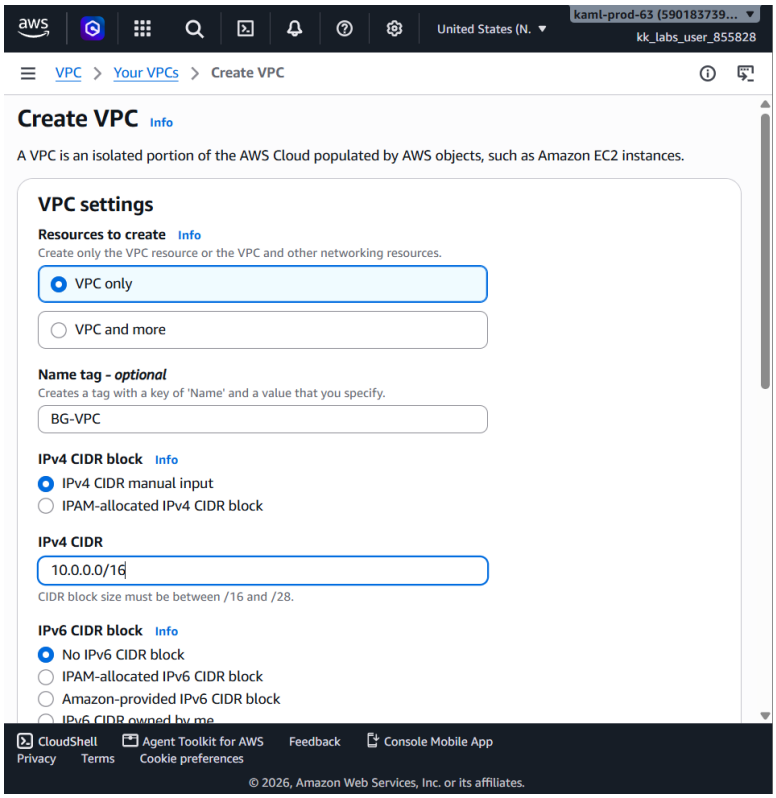
Project Overview

In modern cloud environments, deploying new versions of an application shouldn't mean taking your system offline. This project builds a highly resilient infrastructure by deploying two identical, fully isolated environments (Blue and Green) behind an AWS Application Load Balancer (ALB).

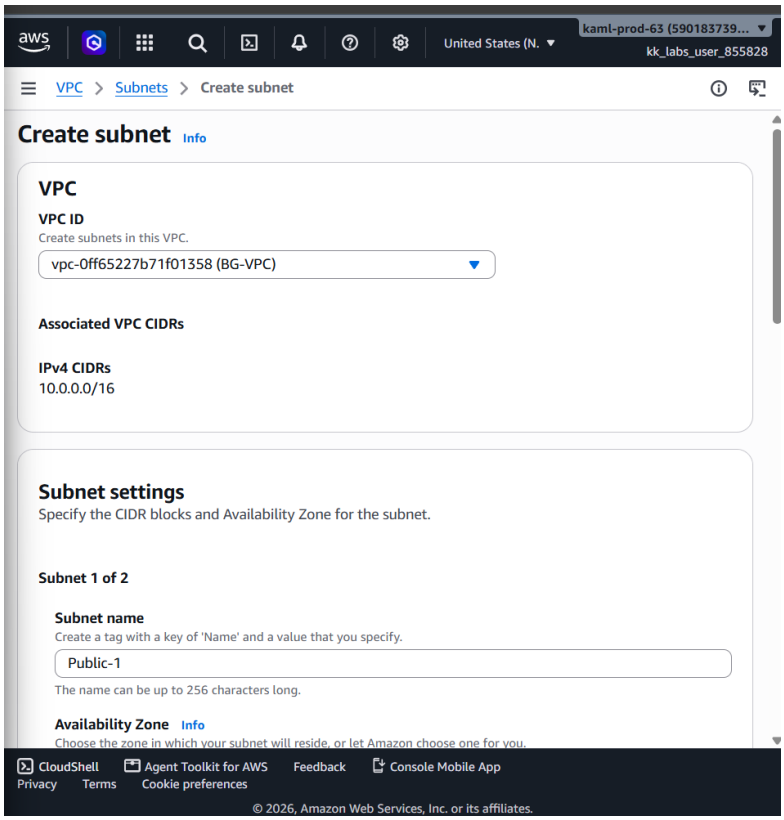
You will learn how to transition traffic seamlessly from your live environment (Version 1 / Blue) to your updated environment (Version 2 / Green) using a Canary deployment strategy. Rather than flipping a switch and hoping for the best, a Canary strategy allows you to route a small percentage of user traffic (e.g., 20%) to the new version to monitor for errors before committing to a full 100% cutover. This ensures zero downtime and minimizes the blast radius if an update fails.

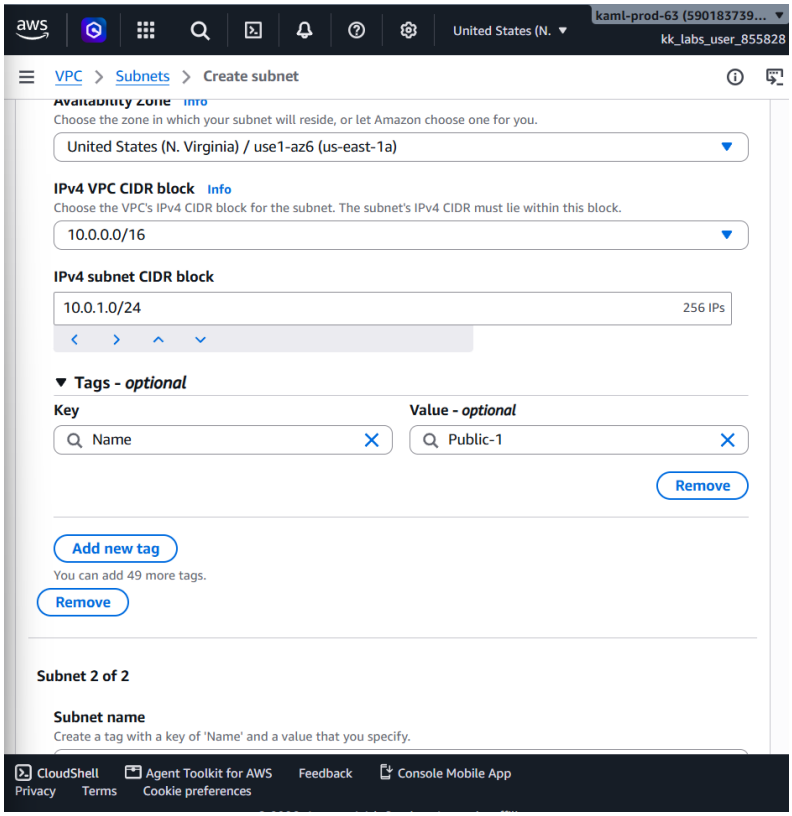


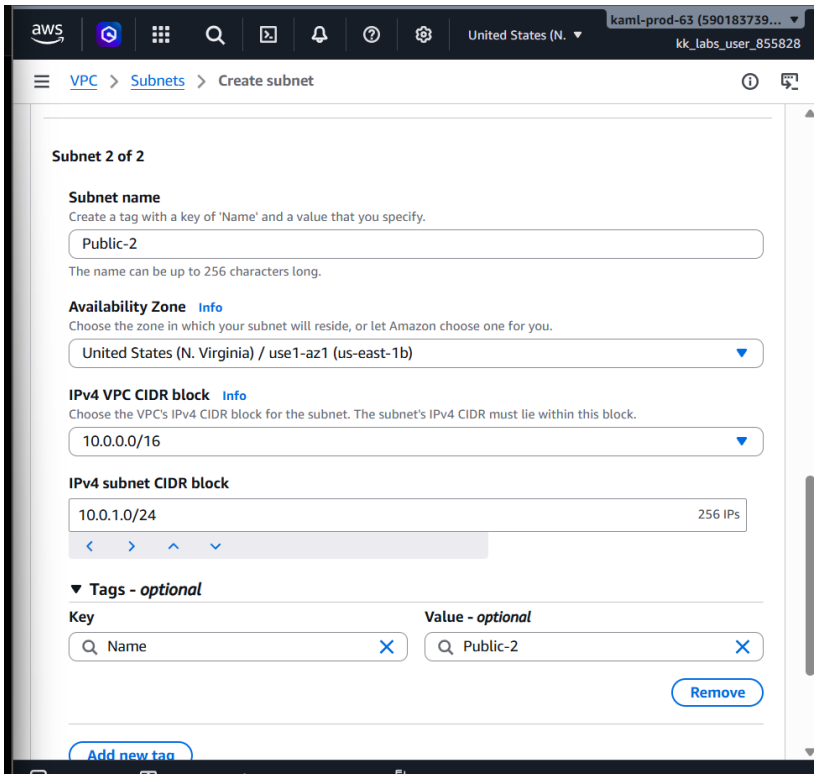
→VPC with atleast 2 AZ’s

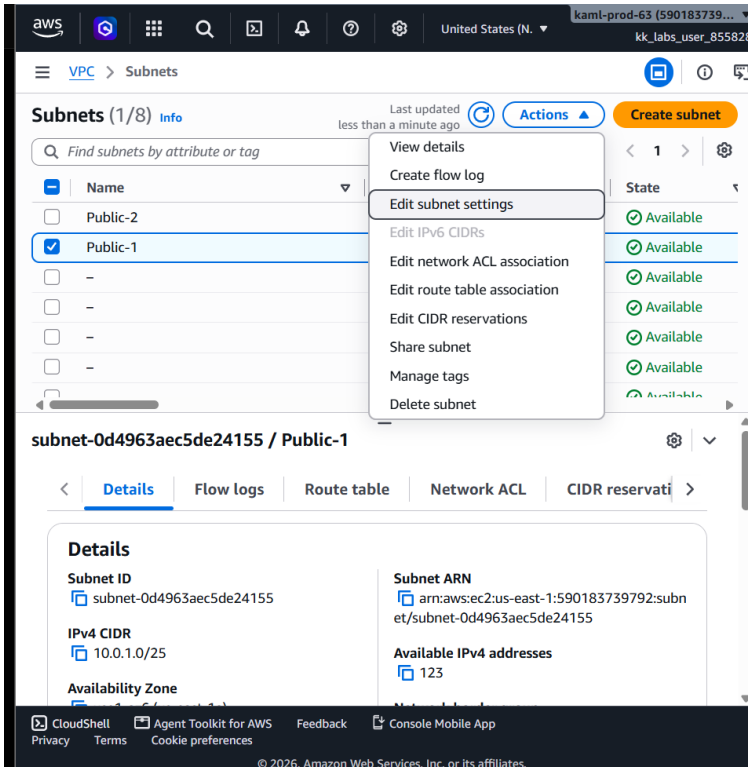


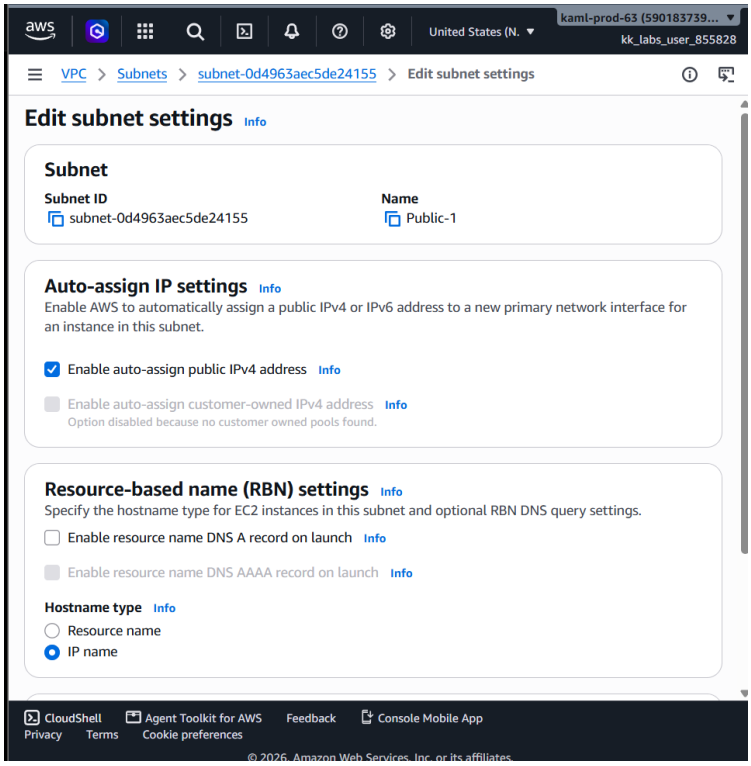
→create 2 public subnets in 2 different AZ’s under same VPC



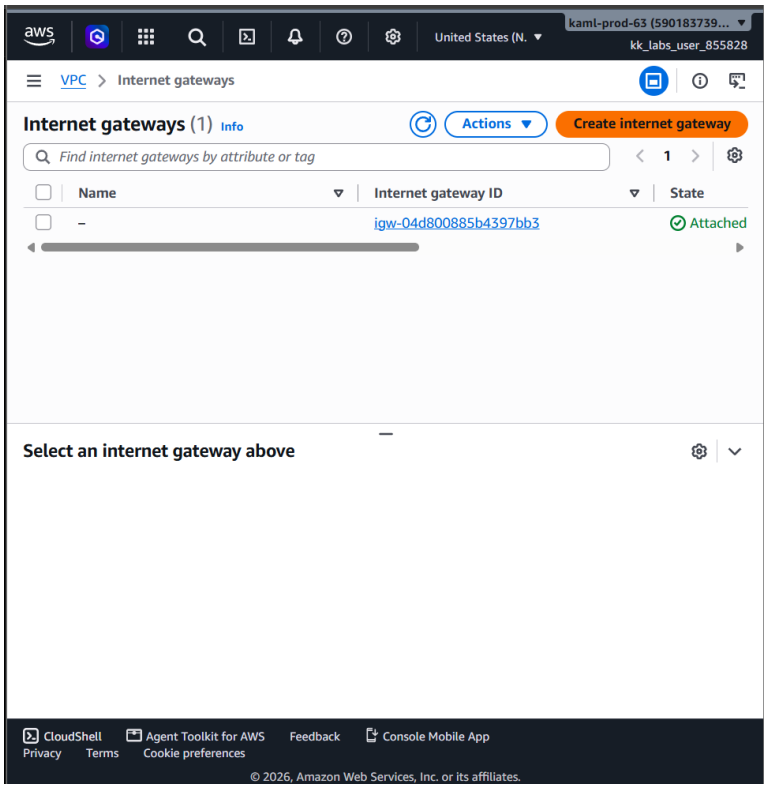

United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
VPC > Subnets > Create subnet
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az6 (us-east-1a)
IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16
IPv4 subnet CIDR block
10.0.1.0/24
256 IPs
Tags - optional
Key
Value - optional
Name
Public-1
Remove
Add new tag
You can add 49 more tags.
Remove
Subnet 2 of 2
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences

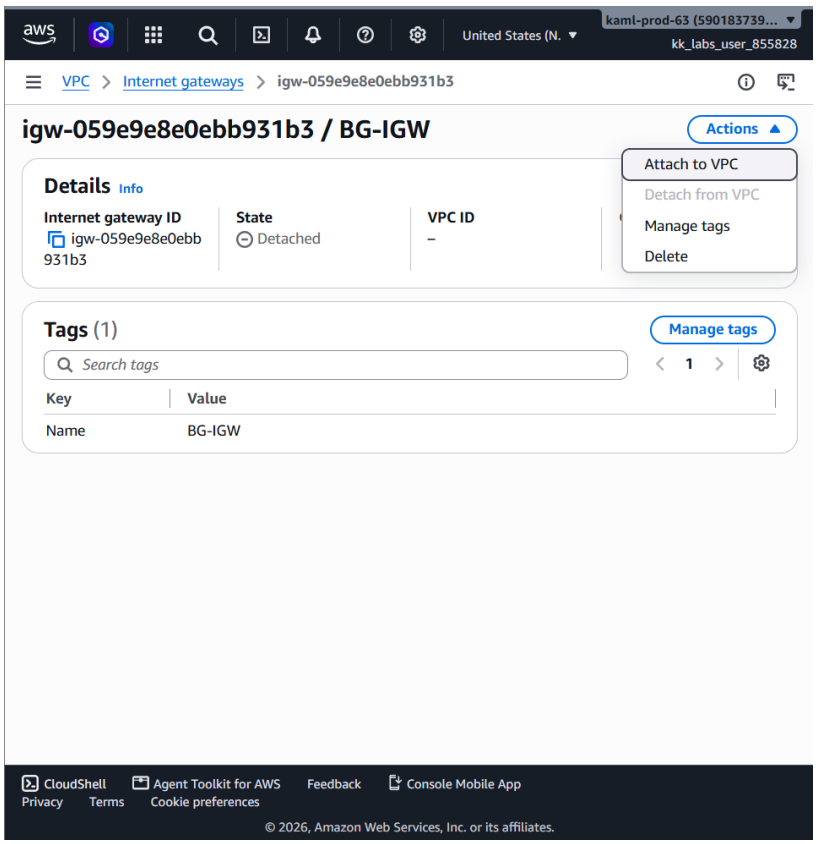

United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
VPC > Subnets > Create subnet
Subnet 2 of 2
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Public-2
The name can be up to 256 characters long.
Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az1 (us-east-1b)
IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16
IPv4 subnet CIDR block
10.0.1.0/24
256 IPs
Tags - optional
Key
Value - optional
Name
Public-2
Remove
Add new tag

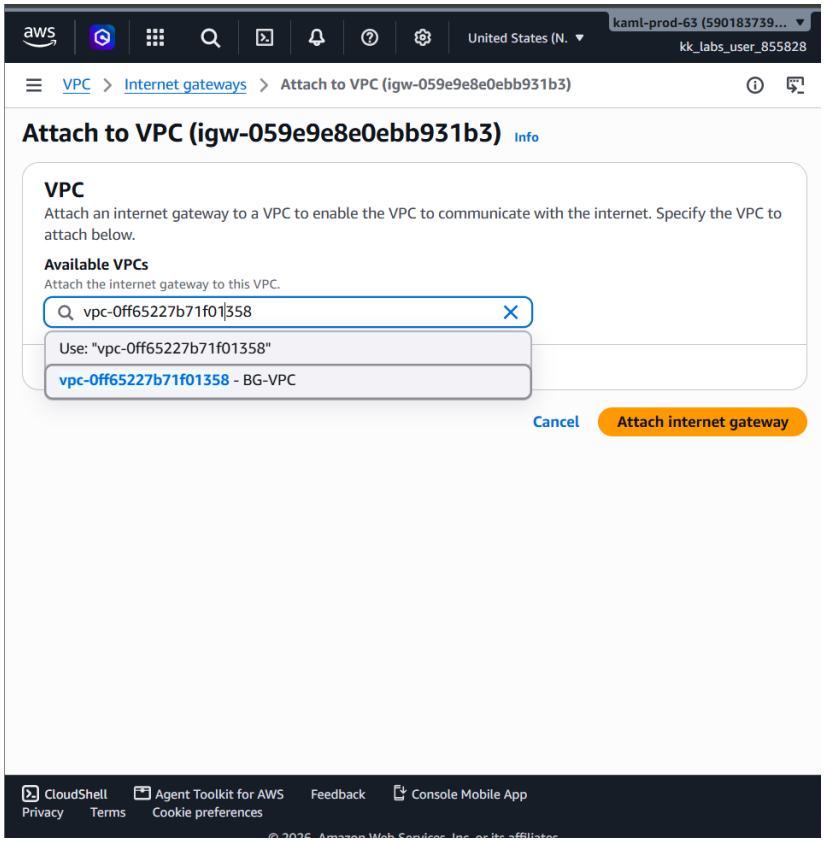

United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
VPC > Subnets
Subnets (1/8) Info
Last updated less than a minute ago
Actions
Create subnet
Find subnets by attribute or tag
1
Name
State
Public-2
Available
Public-1
Available
View details
Create flow log
Edit subnet settings
Edit IPv6 CIDRs
Edit network ACL association
Edit route table association
Edit CIDR reservations
Share subnet
Manage tags
Delete subnet
subnet-0d4963aec5de24155 / Public-1
Details
Flow logs
Route table
Network ACL
CIDR reservati
Details
Subnet ID
subnet-0d4963aec5de24155
Subnet ARN
arn:aws:ec2:us-east-1:590183739792:subnet/subnet-0d4963aec5de24155
IPv4 CIDR
10.0.1.0/25
Available IPv4 addresses
123
Availability Zone
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Cookie preferences
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United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
VPC > Subnets > subnet-0d4963aec5de24155 > Edit subnet settings
Edit subnet settings Info
Subnet
Subnet ID
subnet-0d4963aec5de24155
Name
Public-1
Auto-assign IP settings Info
Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.
Enable auto-assign public IPv4 address Info
Enable auto-assign customer-owned IPv4 address Info
Option disabled because no customer owned pools found.
Resource-based name (RBN) settings Info
Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.
Enable resource name DNS A record on launch Info
Enable resource name DNS AAAA record on launch Info
Hostname type Info
Resource name
IP name
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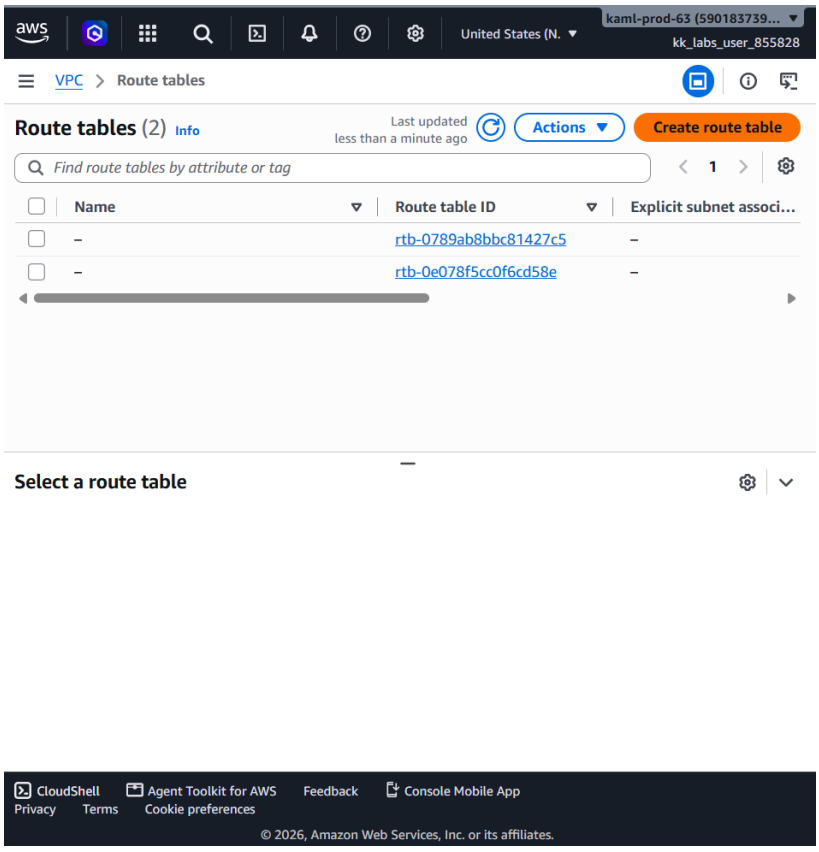
→Create IGW to enable connection to internet for our VPC

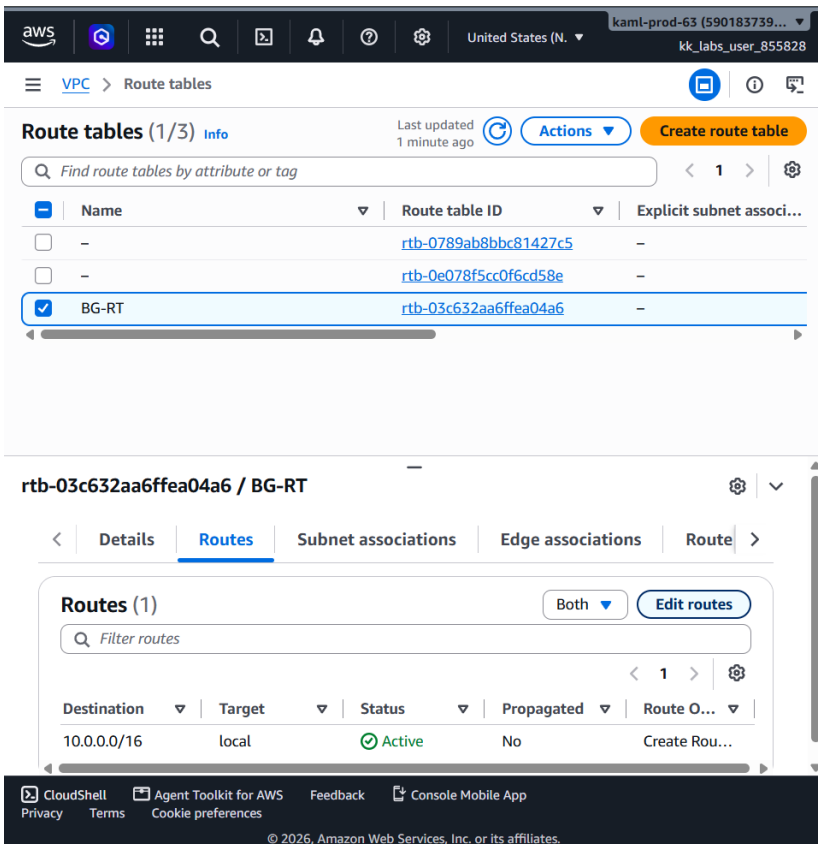




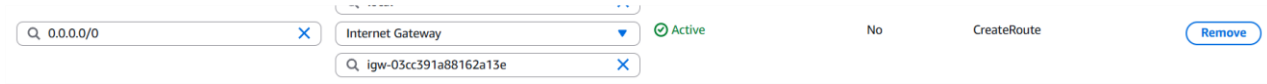


→Create route tables to handle internal traffic

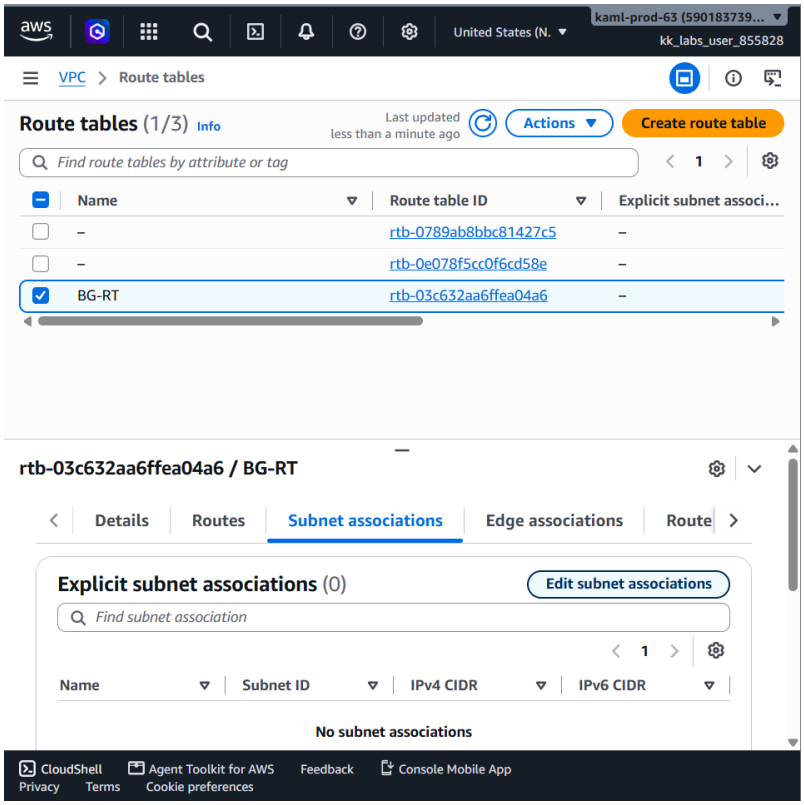


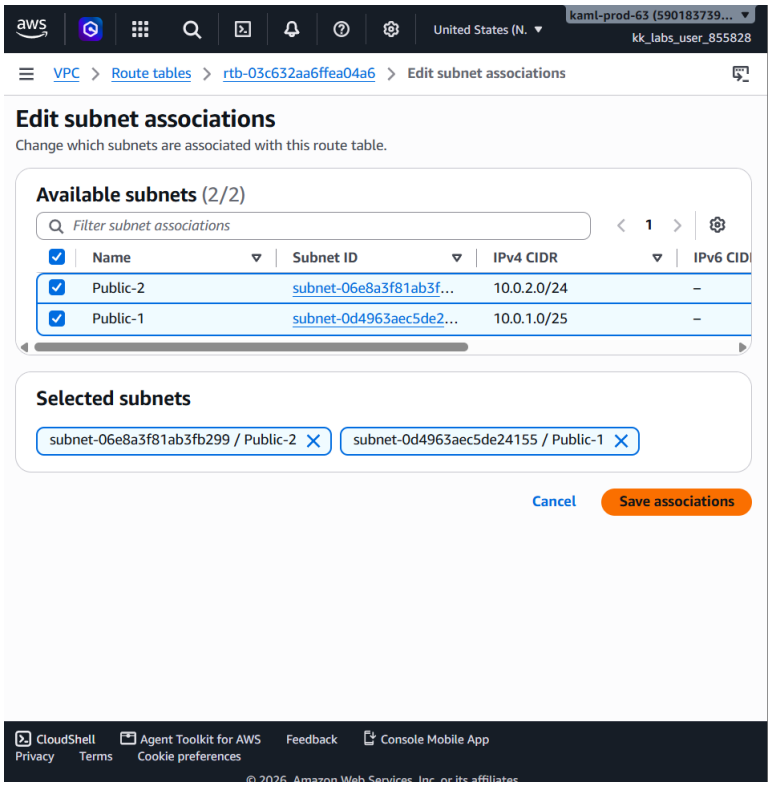


→Edit routes > select internet gate way

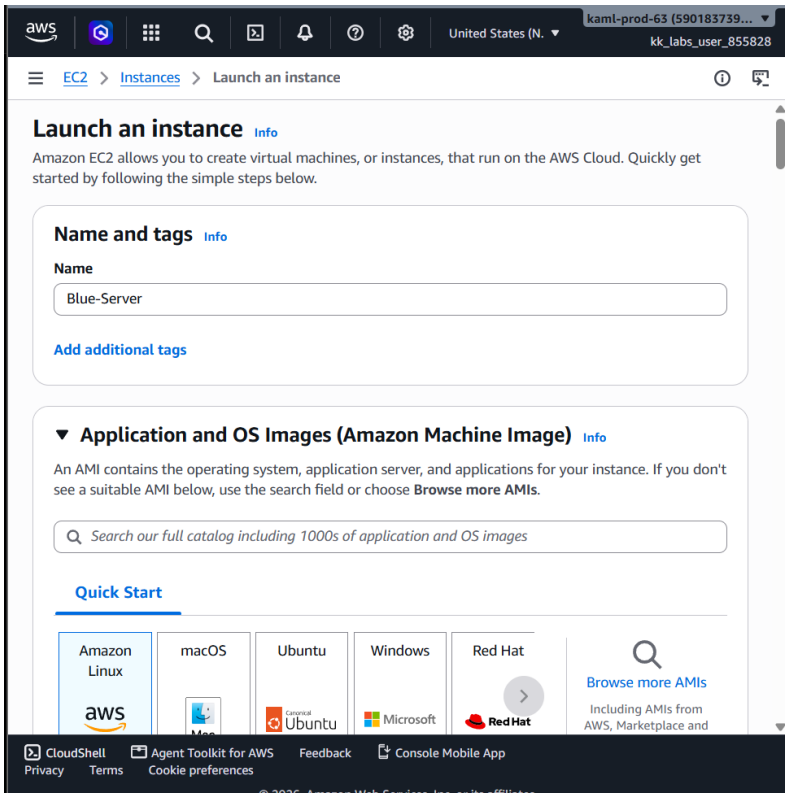


→This enables resources to connnet to internet via IGW

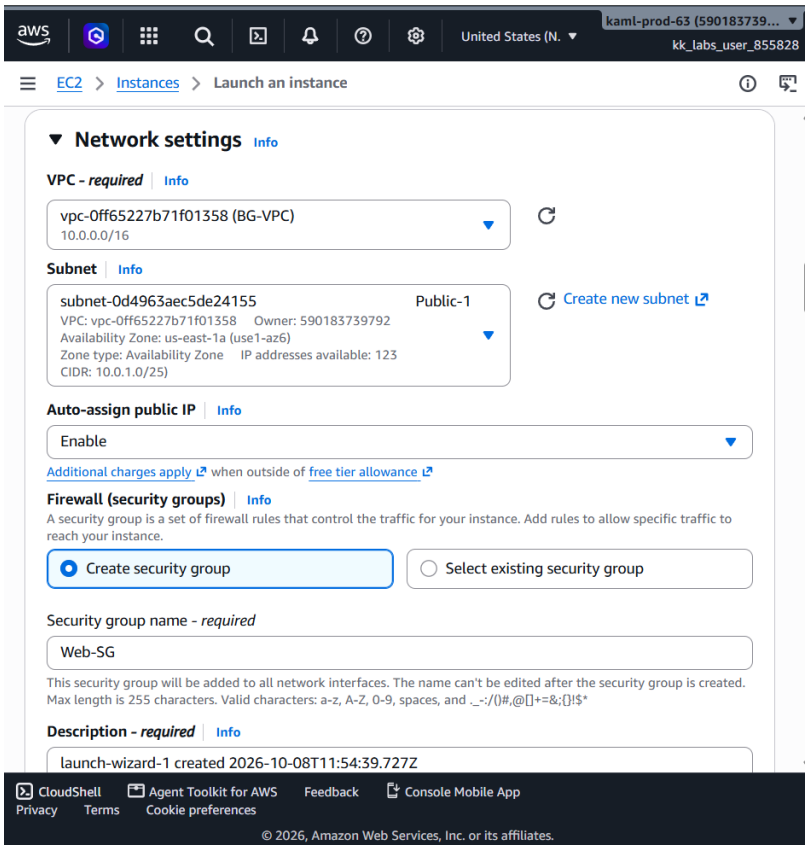

United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
VPC > Route tables
Route tables (1/3) Info
Last updated less than a minute ago
Actions
Create route table
Find route tables by attribute or tag
Name
Route table ID
Explicit subnet associ...
– rtb-0789ab8bbc81427c5 –
– rtb-0e078f5cc0f6cd58e –
BG-RT rtb-03c632aa6ffea04a6 –
rtb-03c632aa6ffea04a6 / BG-RT
Details
Routes
Subnet associations
Edge associations
Route
Explicit subnet associations (0)
Edit subnet associations
Find subnet association
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
No subnet associations
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences


United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
VPC > Route tables > rtb-03c632aa6ffea04a6 > Edit subnet associations
Edit subnet associations
Change which subnets are associated with this route table.
Available subnets (2/2)
Filter subnet associations
Name
Subnet ID
IPv4 CIDR
IPv6 CIDI
Public-2 subnet-06e8a3f81ab3f... 10.0.2.0/24 –
Public-1 subnet-0d4963aec5de2... 10.0.1.0/25 –
Selected subnets
subnet-06e8a3f81ab3fb299 / Public-2
subnet-0d4963aec5de24155 / Public-1
Cancel
Save associations
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences

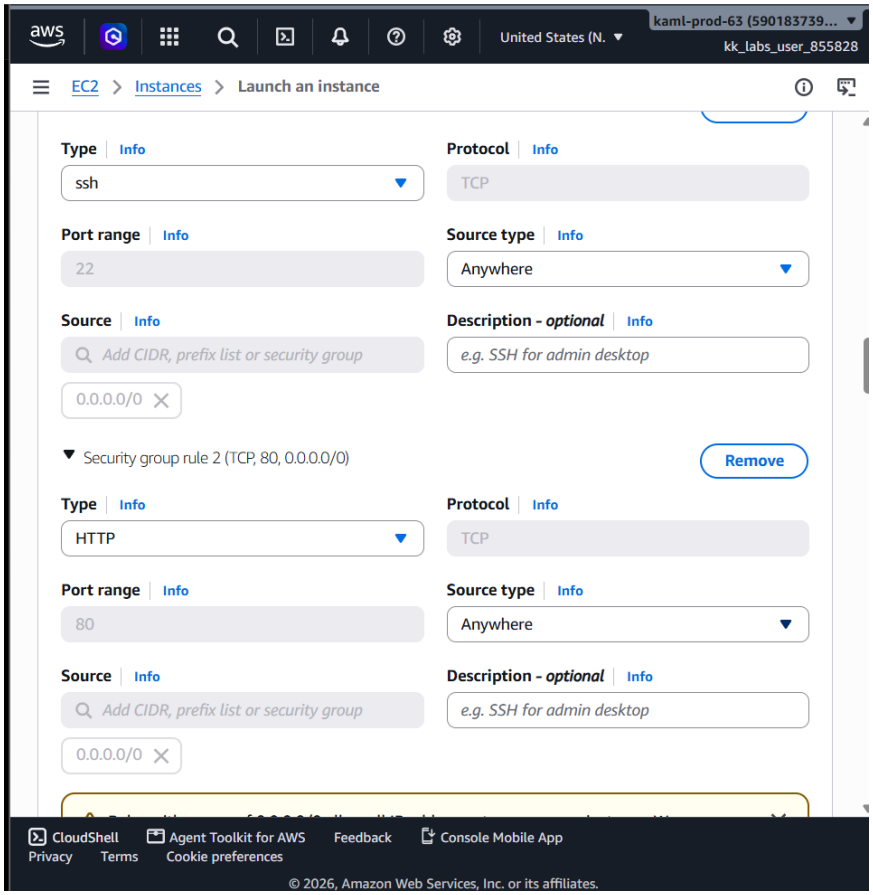
→Main instance -blue

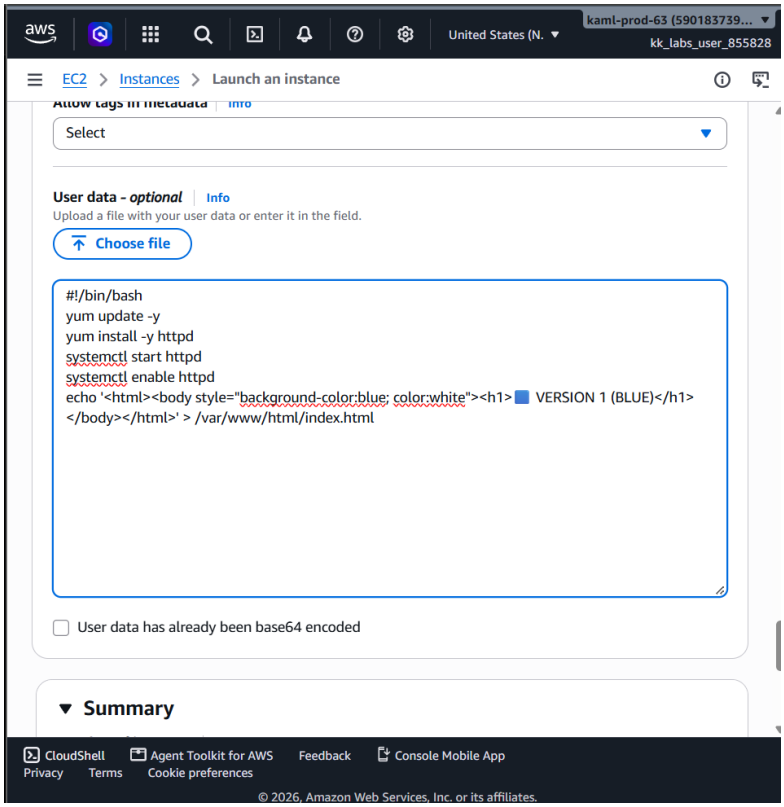


→select subnet public 1

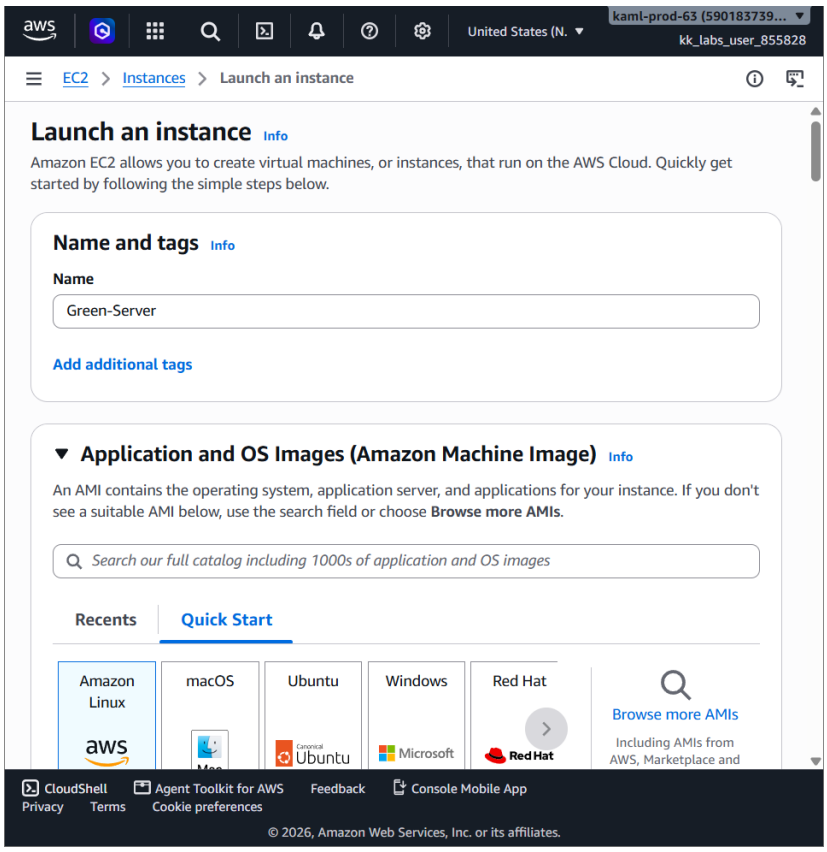


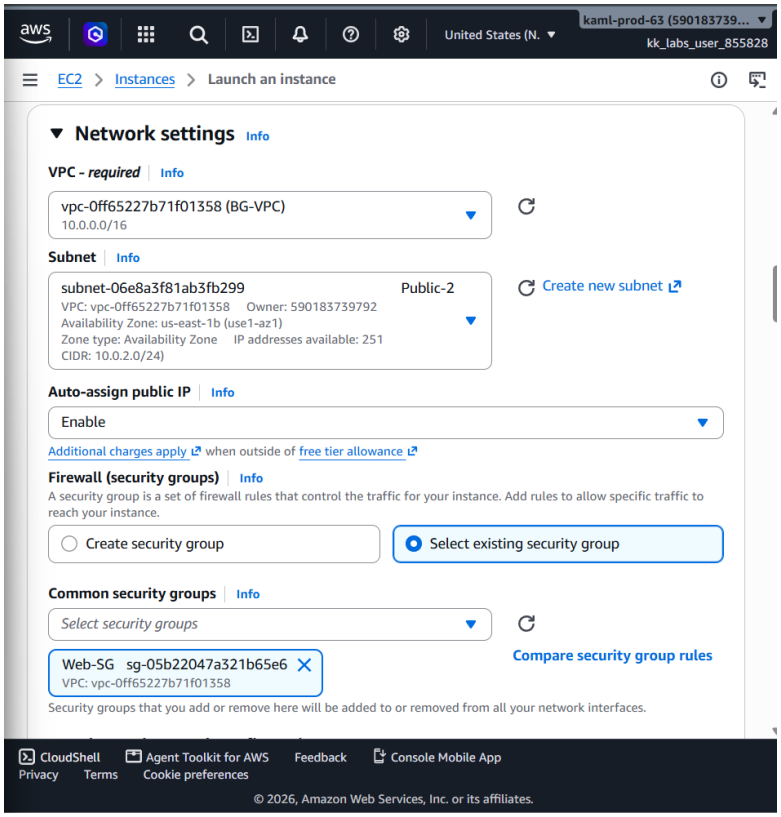
→HTTP to allow http traffic to xccess our resource



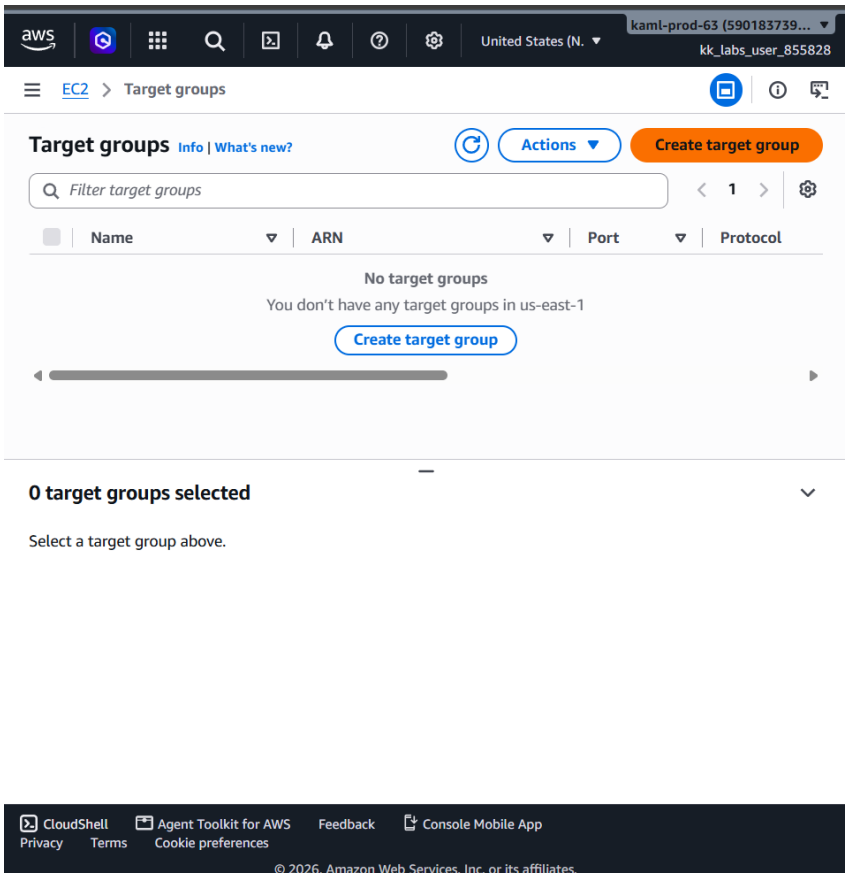


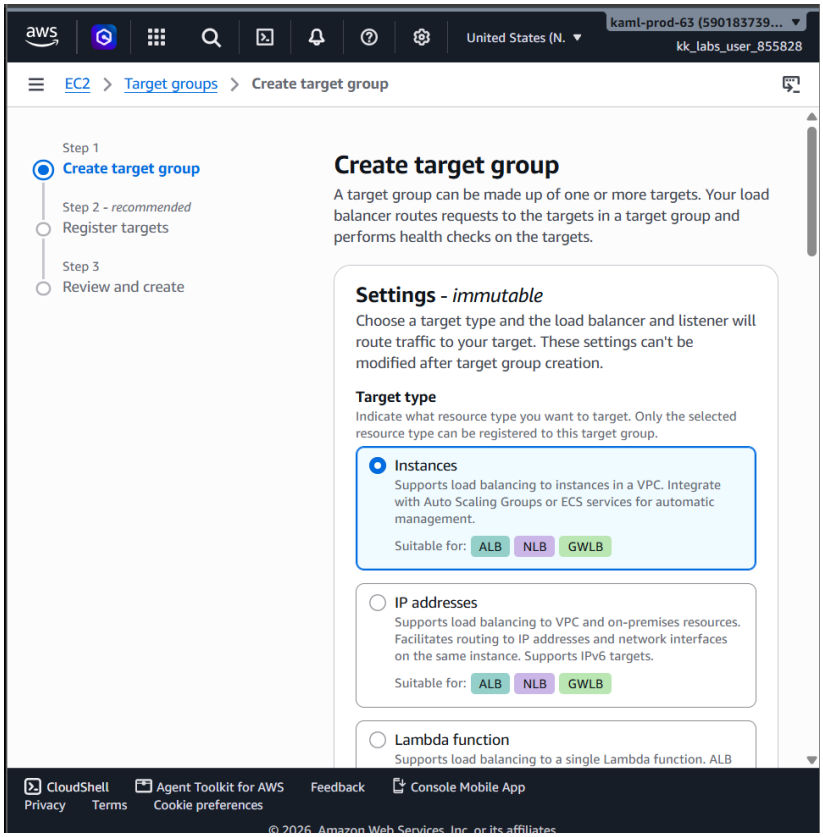
→Second instance- green

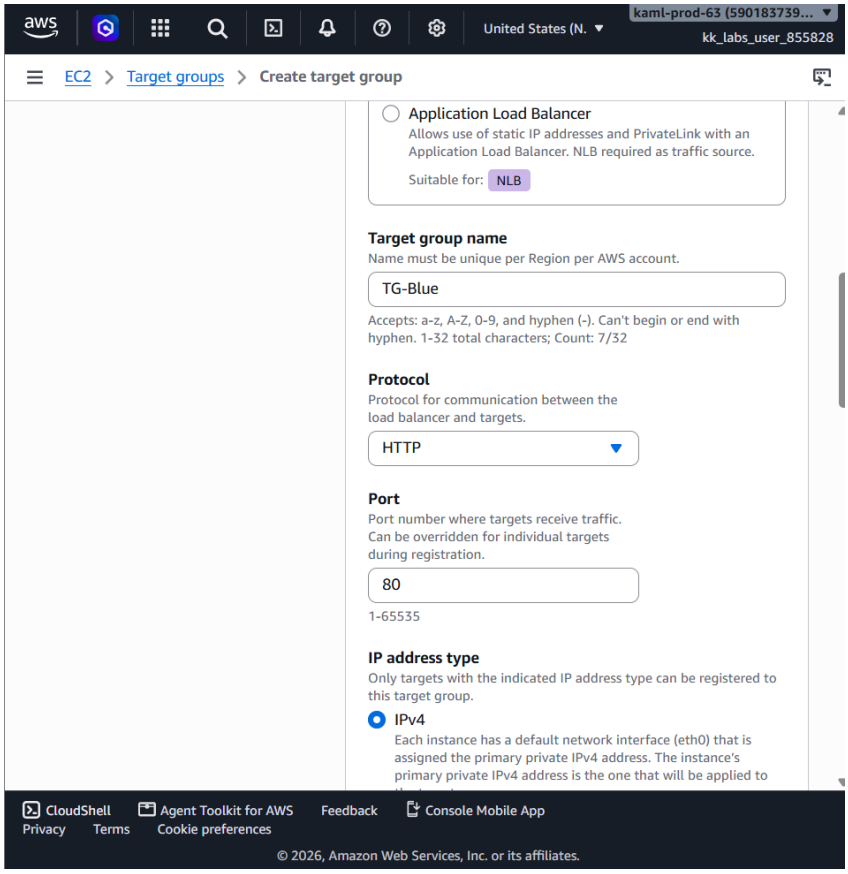


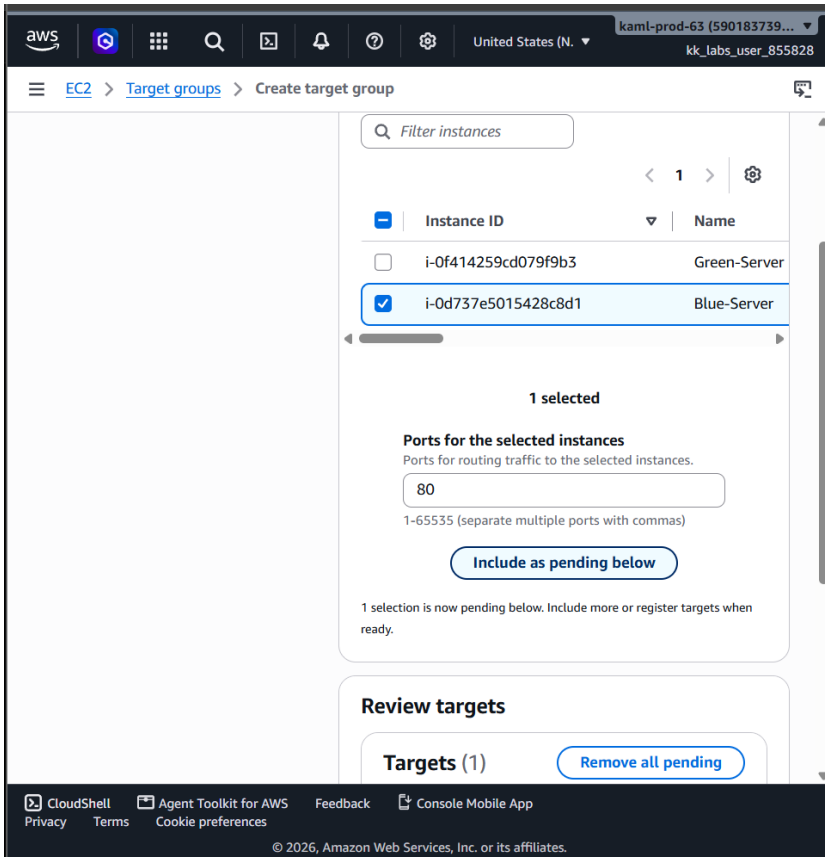


→Target group for Load balancer



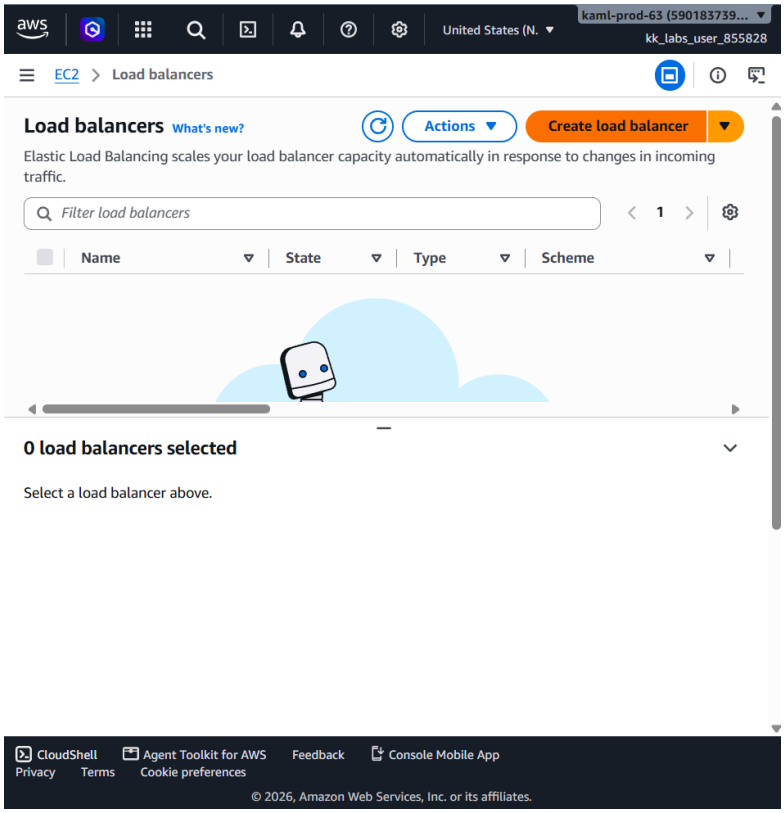

United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
EC2 > Target groups > Create target group
Step 1
Create target group
Step 2 - recommended
Register targets
Step 3
Review and create
Create target group
A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.
Settings - immutable
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.
Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.
Instances
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB
IP addresses
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB
Lambda function
Supports load balancing to a single Lambda function. ALB
CloudShell
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Feedback
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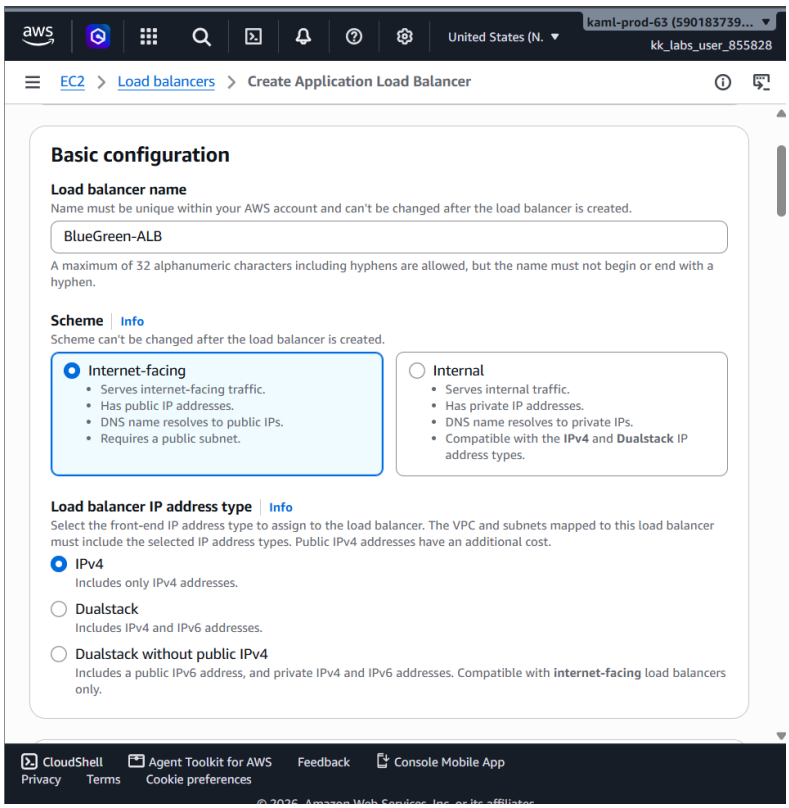

United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
EC2 > Target groups > Create target group
Application Load Balancer
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB
Target group name
Name must be unique per Region per AWS account.
TG-Blue
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 7/32
Protocol
Protocol for communication between the load balancer and targets.
HTTP
Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
80
1-65535
IP address type
Only targets with the indicated IP address type can be registered to this target group.
IPv4
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to
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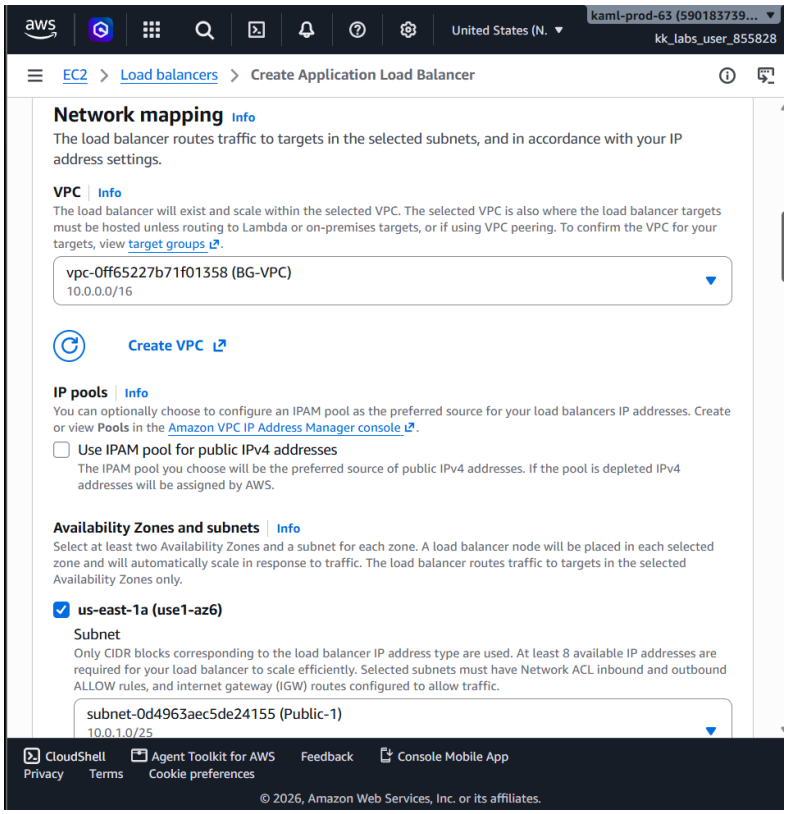


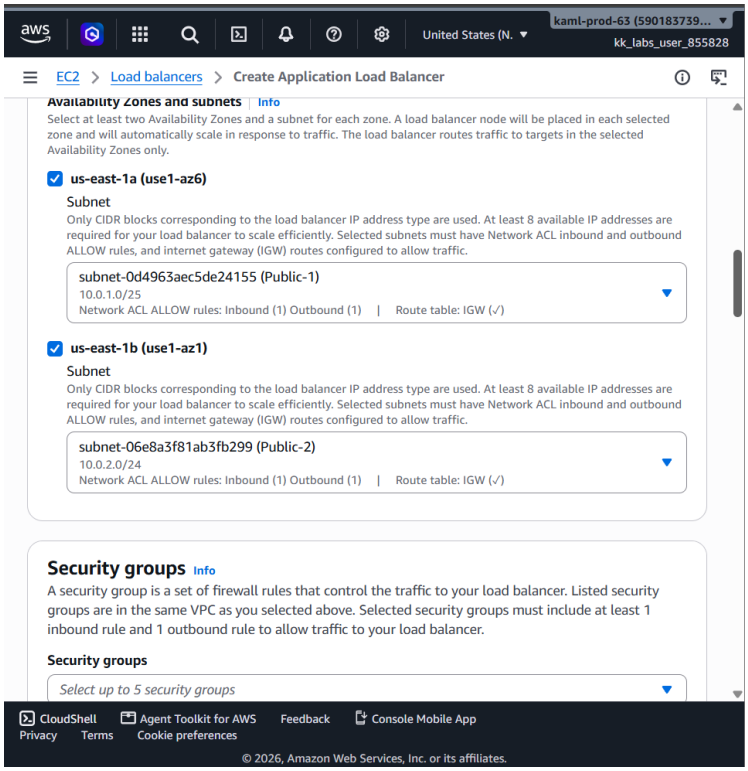
→repeat same procedure for Green instance

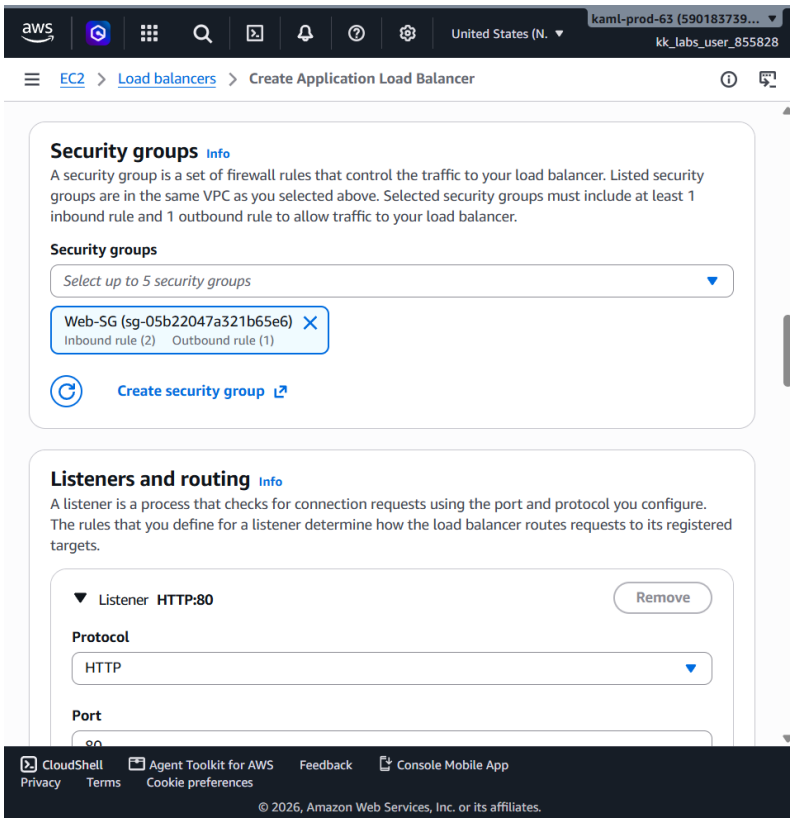
→create load balancer

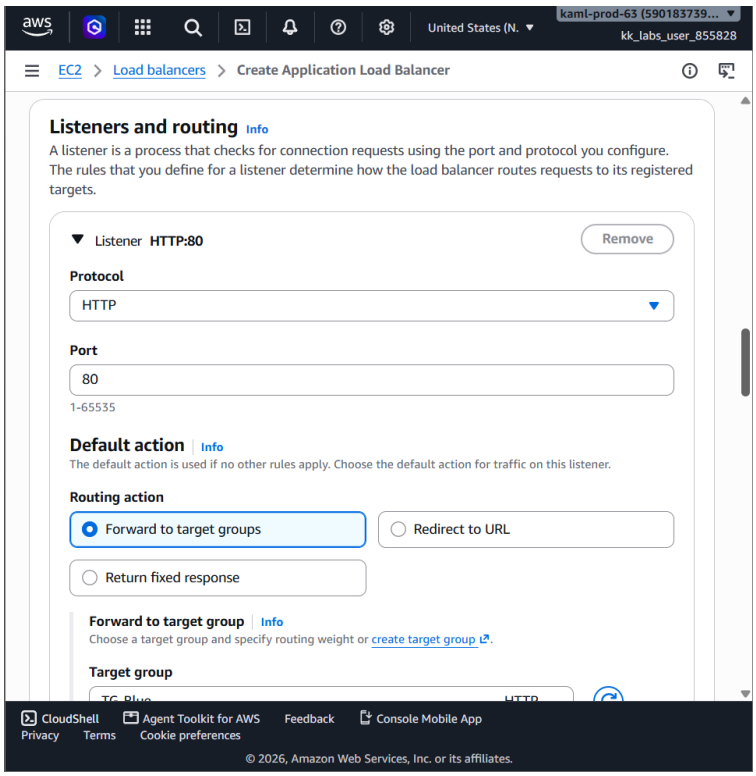



United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
EC2 > Load balancers > Create Application Load Balancer
Basic configuration
Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.
BlueGreen-ALB
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.
Scheme Info
Scheme can't be changed after the load balancer is created.
Internet-facing
Serves internet-facing traffic.
Has public IP addresses.
DNS name resolves to public IPs.
Requires a public subnet.
Internal
Serves internal traffic.
Has private IP addresses.
DNS name resolves to private IPs.
Compatible with the IPv4 and Dualstack IP address types.
Load balancer IP address type Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.
IPv4
Includes only IPv4 addresses.
Dualstack
Includes IPv4 and IPv6 addresses.
Dualstack without public IPv4
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.
CloudShell
Agent Toolkit for AWS
Feedback
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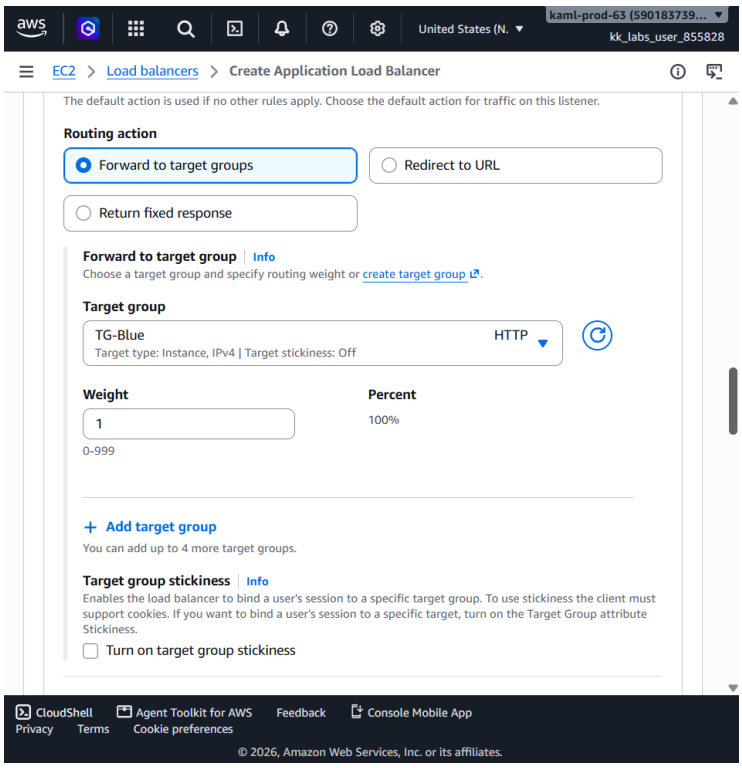

United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
EC2 > Load balancers > Create Application Load Balancer
Network mapping Info
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.
VPC Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups.
vpc-0ff65227b71f01358 (BG-VPC)
10.0.0.0/16
Create VPC
IP pools Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the Amazon VPC IP Address Manager console.
Use IPAM pool for public IPv4 addresses
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.
Availability Zones and subnets Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.
us-east-1a (use1-az6)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.
subnet-0d4963aec5de24155 (Public-1)
10.0.1.0/25
CloudShell
Agent Toolkit for AWS
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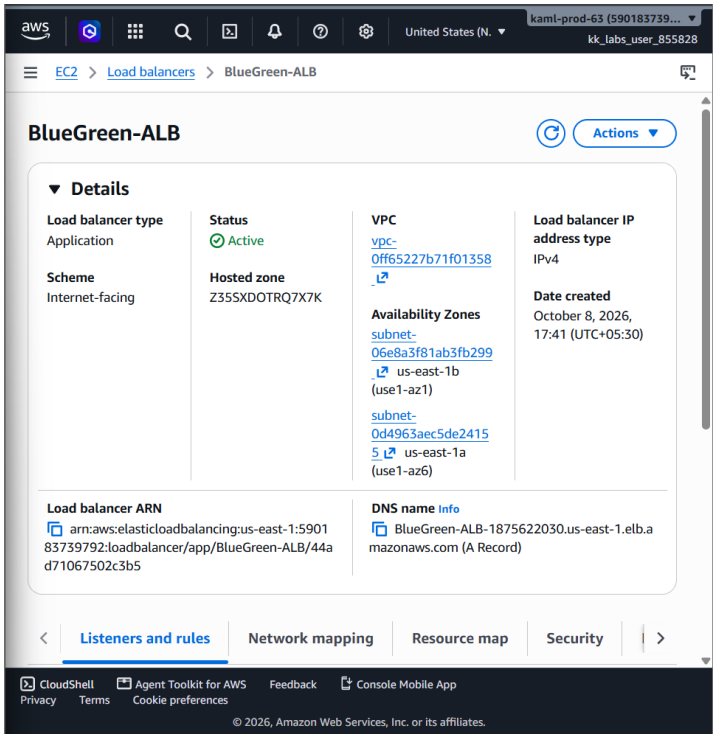

United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
EC2 > Load balancers > Create Application Load Balancer
Availability Zones and subnets Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.
us-east-1a (use1-az6)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.
subnet-0d4963aec5de24155 (Public-1)
10.0.1.0/25
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)
us-east-1b (use1-az1)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.
subnet-06e8a3f81ab3fb299 (Public-2)
10.0.2.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)
Security groups Info
A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.
Security groups
Select up to 5 security groups
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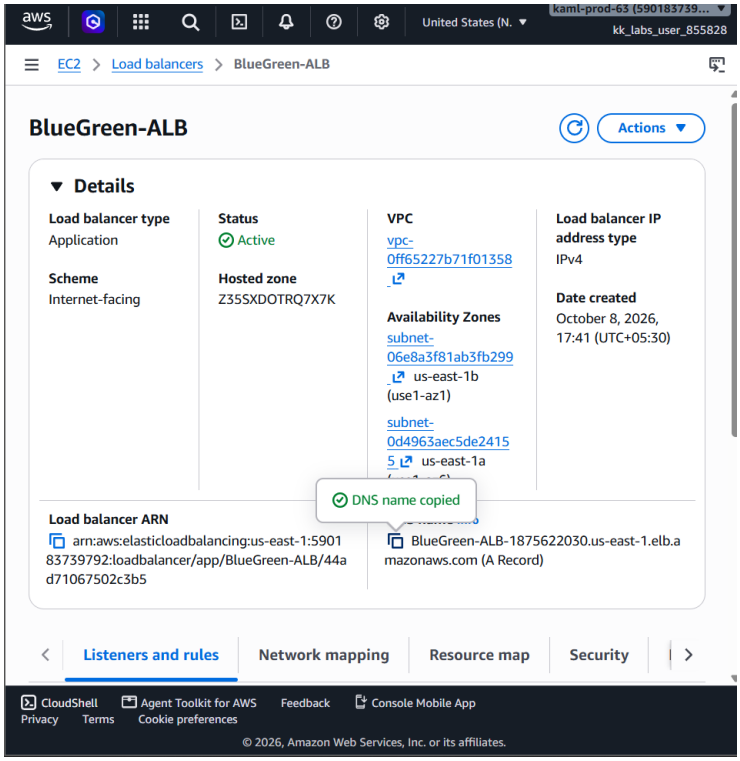

United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
EC2 > Load balancers > Create Application Load Balancer
Security groups Info
A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.
Security groups
Select up to 5 security groups
Web-SG (sg-05b22047a321b65e6)
Inbound rule (2) Outbound rule (1)
Create security group
Listeners and routing Info
A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.
Listener HTTP:80
Remove
Protocol
HTTP
Port
CloudShell
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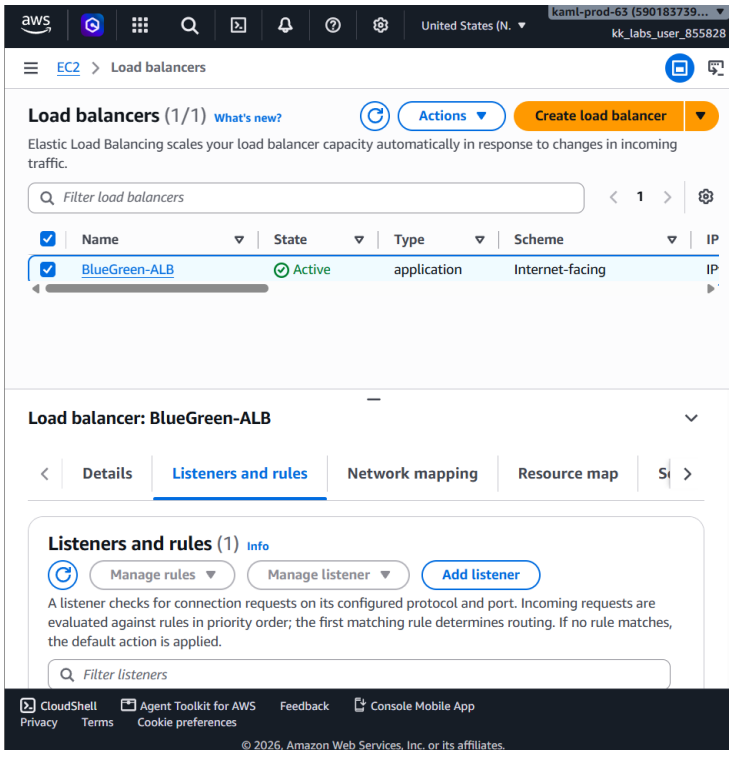
→routing 100% traffice to main/blue server



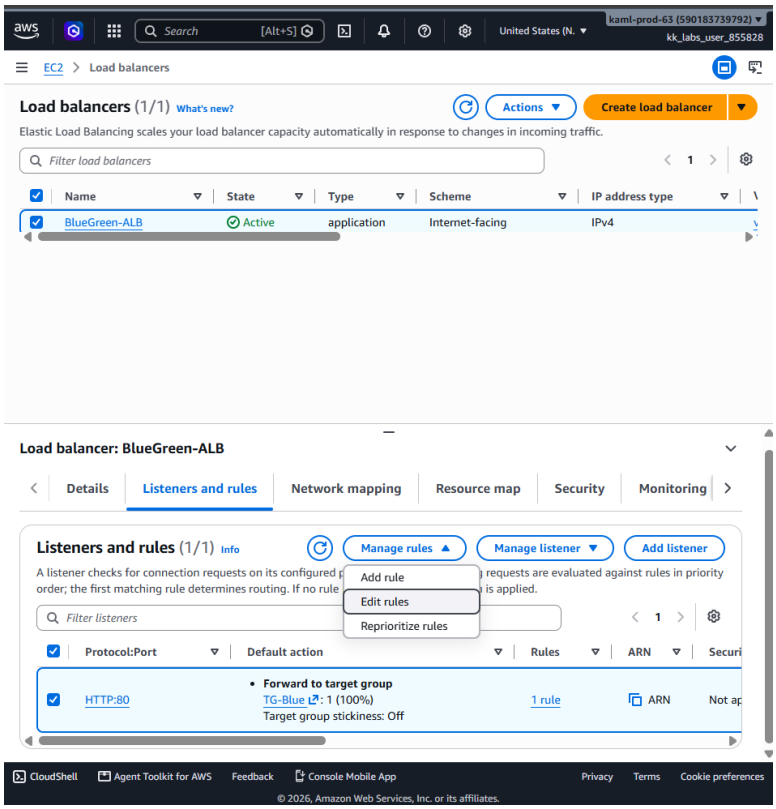

United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
EC2 > Load balancers > BlueGreen-ALB
BlueGreen-ALB
Actions
Details
Load balancer type
Application
Status
Active
VPC
vpc-0ff65227b71f01358
Load balancer IP address type
IPv4
Scheme
Internet-facing
Hosted zone
Z35SXDOTRQ7X7K
Availability Zones
subnet-06e8a3f81ab3fb299 us-east-1b (use1-az1)
subnet-0d4963aec5de24155 us-east-1a (use1-az6)
Date created
October 8, 2026, 17:41 (UTC+05:30)
Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:590183739792:loadbalancer/app/BlueGreen-ALB/44ad71067502c3b5
DNS name Info
BlueGreen-ALB-1875622030.us-east-1.elb.amazonaws.com (A Record)
Listeners and rules
Network mapping
Resource map
Security
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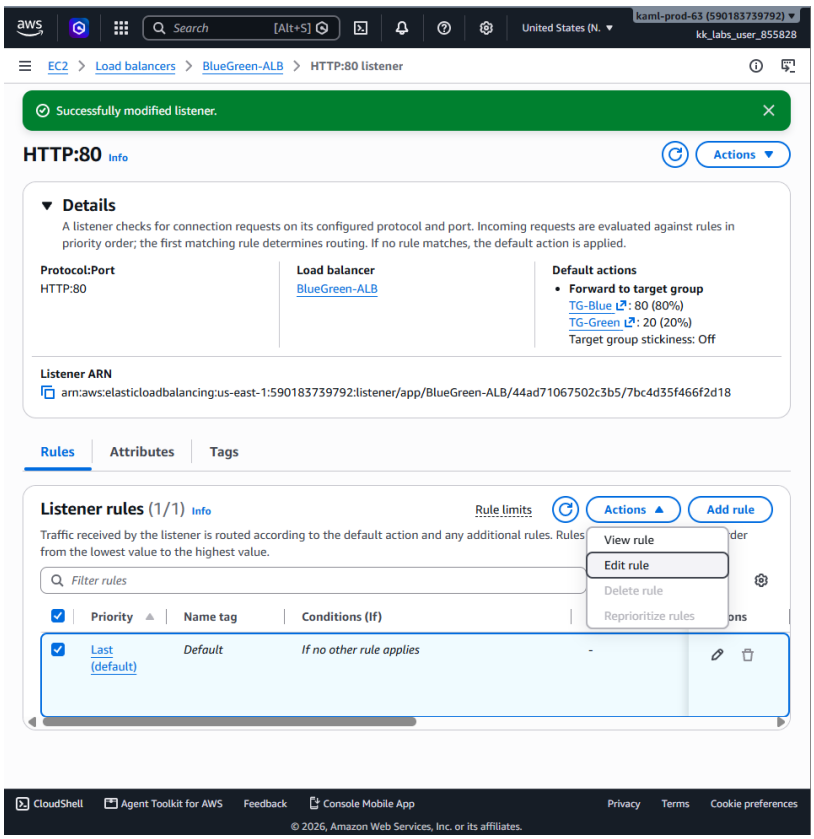

United States (N.
kaml-prod-63 (590183739...
kk_labs_user_855828
EC2 > Load balancers > BlueGreen-ALB
BlueGreen-ALB
Actions
Details
Load balancer type
Application
Status
Active
VPC
vpc-0ff65227b71f01358
Load balancer IP address type
IPv4
Scheme
Internet-facing
Hosted zone
Z35SXDOTRQ7X7K
Availability Zones
subnet-06e8a3f81ab3fb299 us-east-1b (use1-az1)
subnet-0d4963aec5de24155 us-east-1a
Date created
October 8, 2026, 17:41 (UTC+05:30)
DNS name copied
Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:590183739792:loadbalancer/app/BlueGreen-ALB/44ad71067502c3b5
BlueGreen-ALB-1875622030.us-east-1.elb.amazonaws.com (A Record)
Listeners and rules
Network mapping
Resource map
Security
CloudShell
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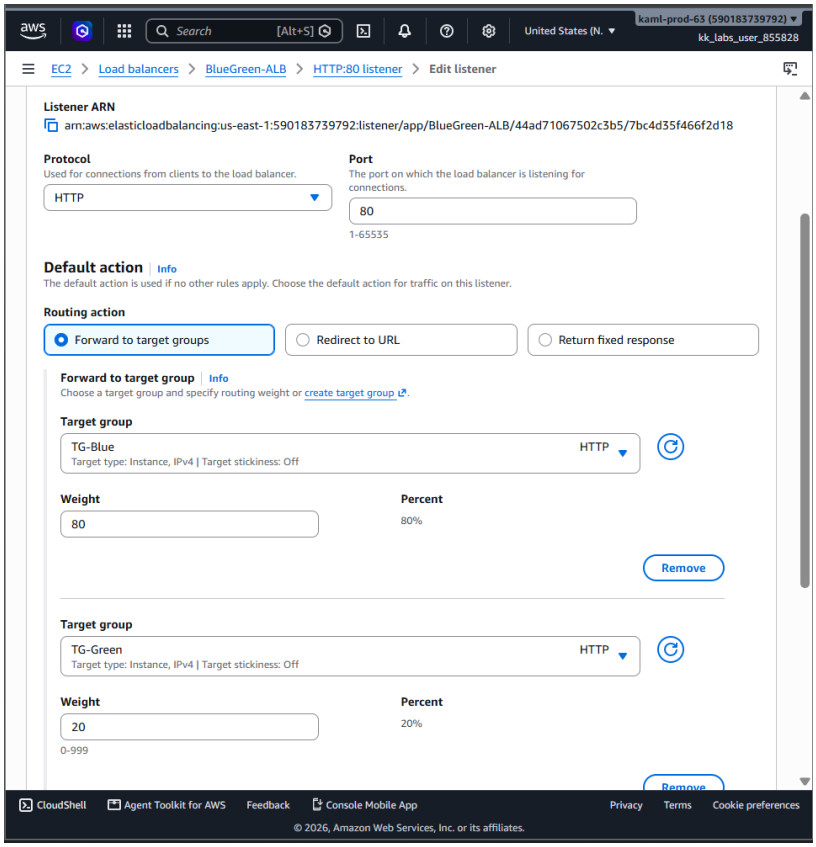

Not secure
bluegreen-alb-1875622030.us-east-1.elb.amazona...
Incognito (2)
Resources
VERSION 1 (BLUE)

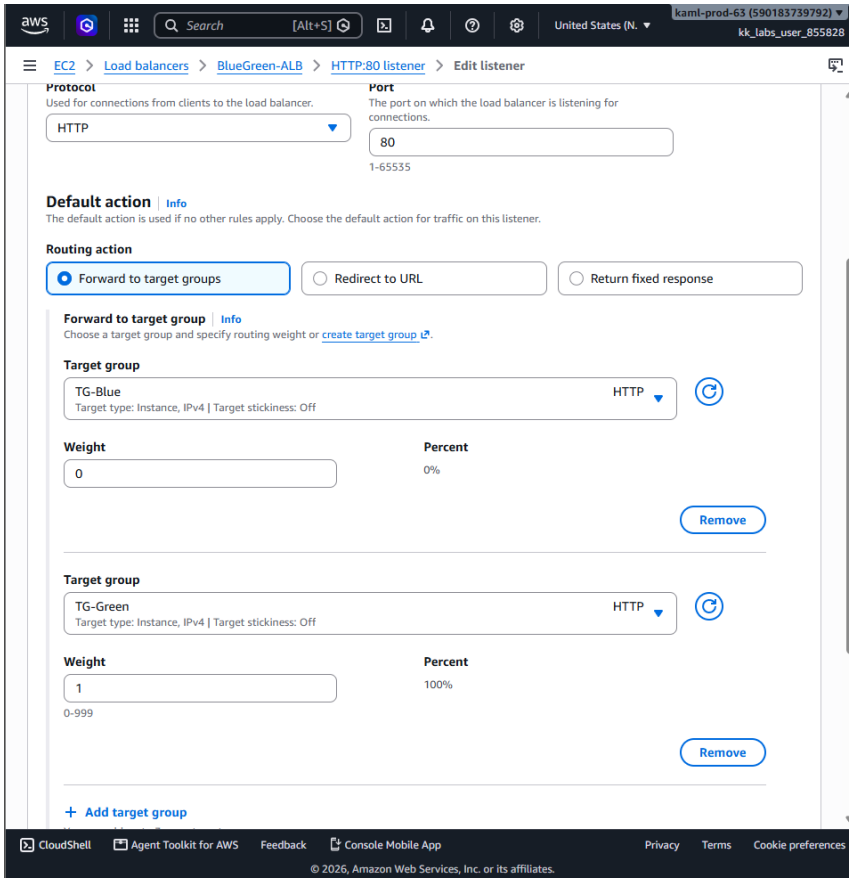

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EC2
Load balancers
Load balancers (1/1)
What's new?
Actions
Create load balancer
Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.
Filter load balancers
Name
State
Type
Scheme
IP
BlueGreen-ALB
Active
application
Internet-facing
Load balancer: BlueGreen-ALB
Details
Listeners and rules
Network mapping
Resource map
Listeners and rules (1)
Info
Manage rules
Manage listener
Add listener
A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.
Filter listeners
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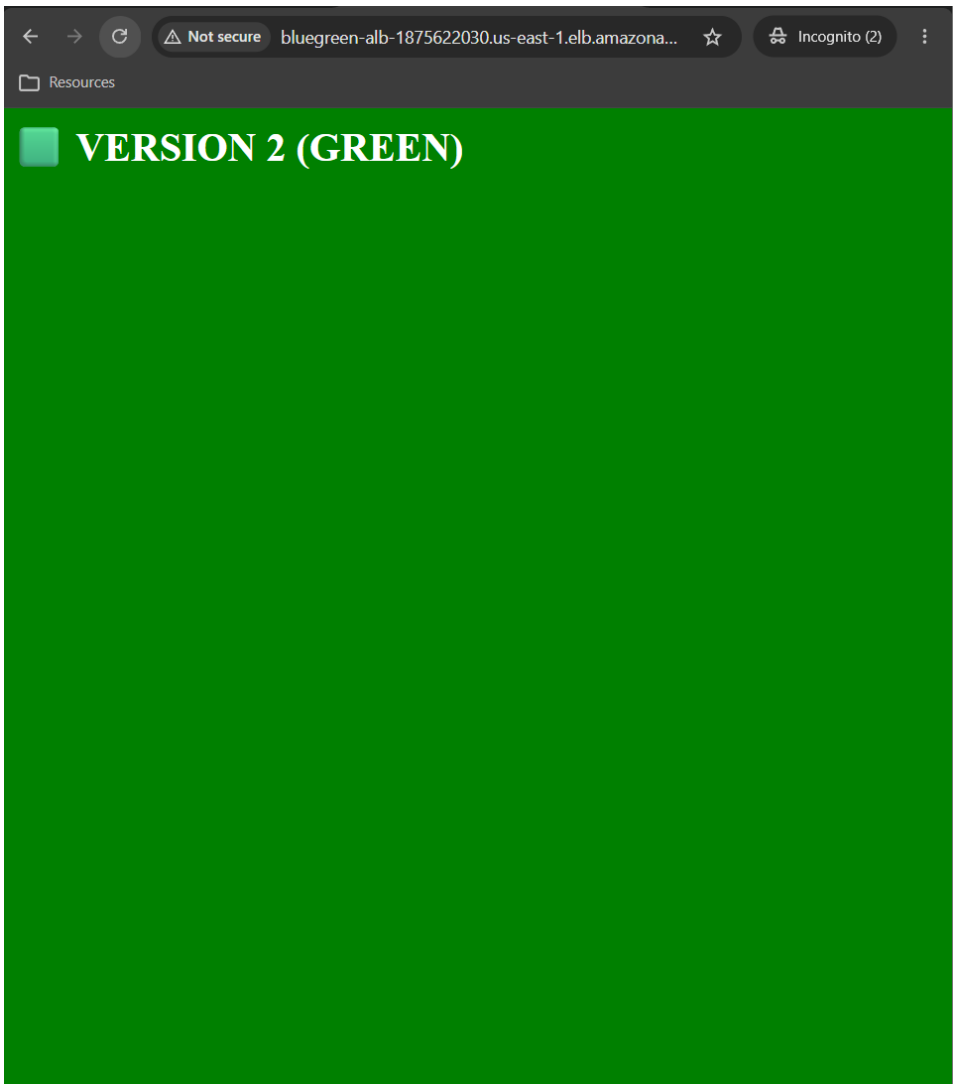
→routing 20% traffic to green server






aws
Search [Alt+S]
United States (N. ▾
kaml-prod-63 (590183739792) ▾
kk_labs_user_855828
EC2 > Load balancers > BlueGreen-ALB > HTTP:80 listener > Edit listener
Listener ARN
arn:aws:elasticloadbalancing:us-east-1:590183739792:listener/app/BlueGreen-ALB/44ad71067502c3b5/7bc4d35f466f2d18
Protocol
Used for connections from clients to the load balancer.
HTTP
Port
The port on which the load balancer is listening for connections.
80
1-65535
Default action | Info
The default action is used if no other rules apply. Choose the default action for traffic on this listener.
Routing action
Forward to target groups
Redirect to URL
Return fixed response
Forward to target group | Info
Choose a target group and specify routing weight or create target group.
Target group
TG-Blue
Target type: Instance, IPv4 | Target stickiness: Off
HTTP
Weight
80
Percent
80%
Remove
Target group
TG-Green
Target type: Instance, IPv4 | Target stickiness: Off
HTTP
Weight
20
0-999
Percent
20%
Remove
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Search [Alt+S]
United States (N. ▾
kaml-prod-63 (590183739792) ▾
kk_labs_user_855828
EC2 > Load balancers > BlueGreen-ALB > HTTP:80 listener > Edit listener
Protocol
Used for connections from clients to the load balancer.
HTTP
Port
The port on which the load balancer is listening for connections.
80
1-65535
Default action | Info
The default action is used if no other rules apply. Choose the default action for traffic on this listener.
Routing action
Forward to target groups
Redirect to URL
Return fixed response
Forward to target group | Info
Choose a target group and specify routing weight or create target group.
Target group
TG-Blue
Target type: Instance, IPv4 | Target stickiness: Off
HTTP
Weight
0
Percent
0%
Remove
Target group
TG-Green
Target type: Instance, IPv4 | Target stickiness: Off
HTTP
Weight
1
0-999
Percent
100%
Remove
+ Add target group
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Not secure
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Incognito (2)
Resources
VERSION 2 (GREEN)